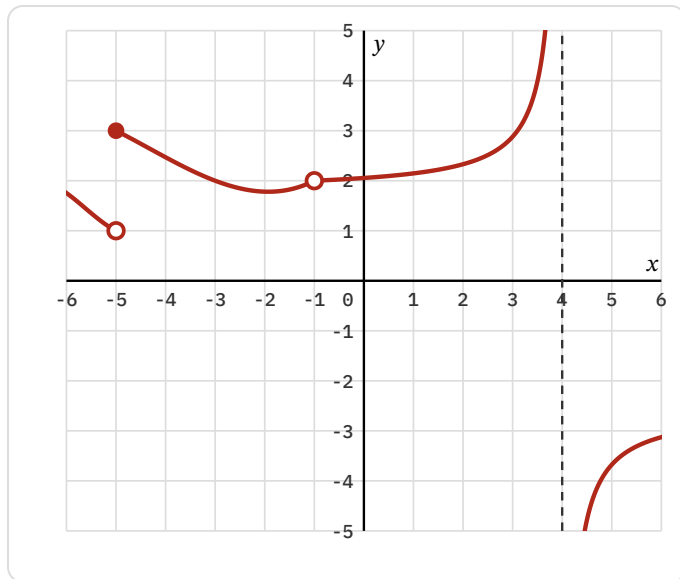


Mock Midterm C — Answer Key

Practice midterm (AI-Free) · Sections 1.1–1.4

Total: 88 points

1. Use the graph of $y = f(x)$ to state the value of each quantity, if it exists. If it does not exist, write DNE. 8 pts



- a.) $f(-1)$ (1)
DNE
- b.) $\lim_{x \rightarrow -1^-} f(x)$ (1)
2
- c.) $\lim_{x \rightarrow -1^+} f(x)$ (1)
2
- d.) $\lim_{x \rightarrow -1} f(x)$ (1)
2
- e.) $f(4)$ (1)
DNE
- f.) $\lim_{x \rightarrow 4^-} f(x)$ (1)
 ∞
- g.) $\lim_{x \rightarrow 4^+} f(x)$ (1)
 $-\infty$
- h.) $\lim_{x \rightarrow 4} f(x)$ (1)
DNE

WORKED SOLUTION

At $x = -1$

There is no filled dot on the line $x = -1$, so $f(-1)$ is undefined $\rightarrow$ DNE.

Tracing the curve toward $x = -1$ from the left, the heights approach 2 (open or closed dot doesn't matter).

Tracing the curve toward $x = -1$ from the right, the heights approach 2 (open or closed dot doesn't matter).

Both one-sided limits equal 2, so $\lim_{x \rightarrow -1} f(x) = 2$ — even though $f(-1)$ is undefined. The limit ignores the value AT the point.

At $x = 4$

There is no filled dot on the line $x = 4$, so $f(4)$ is undefined $\rightarrow$ DNE.

From the left, the curve runs along the asymptote $x = 4$ upward, so the limit is ∞ .

From the right, the curve runs along the asymptote $x = 4$ downward, so the limit is $-\infty$.

The left and right limits don't match, so $\lim_{x \rightarrow 4} f(x)$ DNE.

2. Let $f(x) = \frac{-2}{(x-3)^2}$. Answer with a number, ∞ , $-\infty$, or DNE.

5 pts

a.) $\lim_{x \rightarrow 3^-} f(x)$ (1)

$-\infty$

b.) $\lim_{x \rightarrow 3^+} f(x)$ (1)

$-\infty$

c.) $\lim_{x \rightarrow 3} f(x)$ (1)

$-\infty$

d.) $\lim_{x \rightarrow \infty} f(x)$ (1)

0

e.) The vertical asymptote is $x =$ (1)

3

WORKED SOLUTION

Near $x = 3$ the denominator $(x - 3)^2$ is tiny, so $f(x)$ is huge in size. Only the sign needs work.

Squaring makes the denominator a small **positive** number on both sides, so both one-sided limits are $-\infty$. Since both sides agree we write $\lim_{x \rightarrow 3} f(x) = -\infty$ (it still has no finite value).

As $x \rightarrow \infty$, the denominator grows without bound while the top stays -2, so $f(x) \rightarrow 0$. The vertical asymptote is $x = 3$.

3. Suppose

4 pts

$$\lim_{x \rightarrow 3} f(x) = -3, \quad \lim_{x \rightarrow 3} g(x) = 4, \quad \lim_{x \rightarrow 3} h(x) = 9$$

Use the properties of limits to find each limit.

a.) $\lim_{x \rightarrow 3} x^2 f(x)$ (1)

$$-27$$

b.) $\lim_{x \rightarrow 3} \frac{f(x)g(x)}{h(x)+4}$ (1)

$$-\frac{12}{13}$$

c.) $\lim_{x \rightarrow 3} \sqrt{h(x)} f(x)$ (1)

$$-9$$

d.) $\lim_{x \rightarrow 3} [3f(x) + g(x)]^2$ (1)

$$25$$

WORKED SOLUTION

a. $\lim_{x \rightarrow 3} x^2 f(x) = (3)^2 \cdot (-3) = -27$

b. $\lim_{x \rightarrow 3} \frac{f(x)g(x)}{h(x)+4} = \frac{(-3)(4)}{9+4} = -\frac{12}{13}$

c. $\lim_{x \rightarrow 3} \sqrt{h(x)} f(x) = \sqrt{9} \cdot (-3) = 3 \cdot (-3) = -9$

d. $\lim_{x \rightarrow 3} [3f(x) + g(x)]^2 = [3(-3) + (4)]^2 = (-5)^2 = 25$

The quotient property works here because the bottom limit isn't 0.

4. Calculate each limit algebraically. (Direct substitution gives $\frac{0}{0}$ — simplify first.) If it doesn't exist, write DNE. 4 pts

a.) $\lim_{x \rightarrow 36} \frac{\sqrt{x}-6}{x-36}$ (2)

$\frac{1}{12}$

b.) $\lim_{x \rightarrow -2} \frac{3x^2+4x-4}{x^2+6x+8}$ (2)

-4

WORKED SOLUTION

a.

$$\lim_{x \rightarrow 36} \frac{\sqrt{x}-6}{x-36} = \lim_{x \rightarrow 36} \frac{\sqrt{x}-6}{(\sqrt{x}-6)(\sqrt{x}+6)} = \lim_{x \rightarrow 36} \frac{1}{\sqrt{x}+6} = \frac{1}{6+6} = \frac{1}{12}$$

b.

$$\lim_{x \rightarrow -2} \frac{3x^2+4x-4}{x^2+6x+8} = \lim_{x \rightarrow -2} \frac{(x+2)(3x-2)}{(x+2)(x+4)} = \lim_{x \rightarrow -2} \frac{3x-2}{x+4} = \frac{-8}{2} = -4$$

Notation tip: keep writing **lim** on every line until the step where you actually substitute.

5. Compute each limit algebraically. Use ∞ / $-\infty$ for unbounded results.

6 pts

a.) $\lim_{x \rightarrow \infty} \frac{-3x+7}{-2x^2-9x}$ (2)
 0

b.) $\lim_{x \rightarrow -\infty} \frac{-x^3}{6x^2+6x}$ (2)
 ∞

c.) $\lim_{x \rightarrow \infty} \frac{4x^3+6x^2+6x}{-5x^3-8x^2+1}$ (2)
 $-\frac{4}{5}$

WORKED SOLUTION

a. Divide every term by x^2 , the highest power of x in the denominator:

$$\lim_{x \rightarrow \infty} \frac{-3x+7}{-2x^2-9x} = \lim_{x \rightarrow \infty} \frac{-\frac{3}{x} + \frac{7}{x^2}}{-2 - \frac{9}{x}}$$

The top $\rightarrow 0$ while the bottom $\rightarrow -2$, so the limit is 0 .

b. Divide every term by x^2 , the highest power of x in the denominator:

$$\lim_{x \rightarrow -\infty} \frac{-x^3}{6x^2+6x} = \lim_{x \rightarrow -\infty} \frac{-x}{6 + \frac{6}{x}}$$

The bottom $\rightarrow 6$, but the top is dominated by $-1x^1$, which $\rightarrow \infty$ as $x \rightarrow -\infty$; dividing by 6 gives ∞ .

c. Divide every term by x^3 , the highest power of x in the denominator:

$$\lim_{x \rightarrow \infty} \frac{4x^3+6x^2+6x}{-5x^3-8x^2+1} = \lim_{x \rightarrow \infty} \frac{4 + \frac{6}{x} + \frac{6}{x^2}}{-5 - \frac{8}{x} + \frac{1}{x^3}}$$

Every fraction term $\rightarrow 0$, leaving $\frac{4}{-5} = -\frac{4}{5}$.

6. Find each limit (end behavior). Use ∞ or $-\infty$ when the outputs grow without bound.

4 pts

a.) $\lim_{x \rightarrow -\infty} (-4x^5 + 5x^2 + 3x)$ (1)
 ∞

b.) $\lim_{x \rightarrow -\infty} (-2x^5 - 3x^4 + x)$ (1)
 ∞

c.) $\lim_{x \rightarrow -\infty} (-3 - \frac{8}{x^3})$ (1)
 -3

d.) $\lim_{x \rightarrow -\infty} (-e^{-x})$ (1)
 $-\infty$

WORKED SOLUTION

a. Leading term $-4x^5$: degree 5 (odd), leading coefficient negative. As $x \rightarrow -\infty$, x^5 is negative, times -4 $\rightarrow \infty$.

b. Leading term $-2x^5$: degree 5 (odd), leading coefficient negative. As $x \rightarrow -\infty$, x^5 is negative, times -2 $\rightarrow \infty$.

c. $\frac{8}{x^3} \rightarrow 0$ as $x \rightarrow -\infty$, leaving -3.

d. As $x \rightarrow -\infty$, e^{-x} grows without bound, and multiplying by -1 gives $-\infty$.

7. Let

6 pts

$$f(x) = \begin{cases} \frac{2x^2 + 15x + 25}{2x + 5} & x < -\frac{5}{2} \\ -2x - 3 & x \geq -\frac{5}{2} \end{cases}$$

Find each limit or function value algebraically. Write DNE if it doesn't exist.

a.) $\lim_{x \rightarrow 1} f(x)$ (1)

-5

b.) $\lim_{x \rightarrow -5} f(x)$ (1)

0

c.) $f\left(-\frac{5}{2}\right)$ (1)

2

d.) $\lim_{x \rightarrow -\frac{5}{2}^-} f(x)$ (1)

$\frac{5}{2}$

e.) $\lim_{x \rightarrow -\frac{5}{2}^+} f(x)$ (1)

2

f.) $\lim_{x \rightarrow -\frac{5}{2}} f(x)$ (1)

DNE

WORKED SOLUTION

For $x < -\frac{5}{2}$, factor the top: $\frac{(2x+5)(x+5)}{2x+5} = x + 5$ (valid since $x \neq -\frac{5}{2}$ there).

a. $1 > -\frac{5}{2} \rightarrow$ bottom piece: $-2(1) - 3 = -5$.

b. $-5 < -\frac{5}{2} \rightarrow$ top piece, simplified: $-5 + 5 = 0$.

c. The bottom piece includes $x = -\frac{5}{2}$ ($\geq$), so $f\left(-\frac{5}{2}\right) = -2\left(-\frac{5}{2}\right) - 3 = 2$.

d. Left: $\lim_{x \rightarrow -\frac{5}{2}^-} (x + 5) = \frac{5}{2}$.

e. Right: $\lim_{x \rightarrow -\frac{5}{2}^+} (-2x - 3) = 2$.

f. $\frac{5}{2} \neq 2$, so the limit DNE.

8. Find the value of k that makes f continuous at $x = 1$.

4 pts

$$f(x) = \begin{cases} x^2 - x & x < 1 \\ kx - 6 & x \geq 1 \end{cases}$$

a.) $\lim_{x \rightarrow 1^-} f(x)$ (2)
0

b.) $k =$ (2)
6

WORKED SOLUTION

Left: $(1)^2 - 1(1) = 0$. Right (and $f(1)$): $k(1) - 6$.

$$1k - 6 = 0 \Rightarrow k = \frac{6}{1} = 6$$

9. Classify each function at the given x -value.

3 pts

a.) $f(x) = \begin{cases} x - 2 & x < 2 \\ x - 4 & x \geq 2 \end{cases}$ at $x = 2$ (1)
(C) Jump discontinuity

b.) $f(x) = \frac{x^2 - 16}{x - 4}$ at $x = 4$ (1)
(B) Removable discontinuity

c.) $f(x) = \frac{x + 2}{x^2 + 6x + 8}$ at $x = -4$ (1)
(D) Infinite discontinuity

WORKED SOLUTION

a. Left $\rightarrow 0$, right $\rightarrow -2$. Different finite values $\rightarrow$ jump.

b. $0/0$ at 4, but it cancels to $x + 4$, so the limit exists (8) while $f(4)$ is undefined $\rightarrow$ removable.

c. Factor: $\frac{(x+2)}{(x+2)(x+4)}$. The $(x + 4)$ factor does NOT cancel, so $x = -4$ is a vertical asymptote $\rightarrow$ infinite.

10. Let $f(x) = x^2 + 6x + 5$, where $f(x)$ represents the number of miles driven after x hours. 7 pts
Find the average rate of change over $[2, 6]$.

a.) $f(2)$ (2)

21

b.) $f(6)$ (2)

77

c.) Average rate of change (number only) (2)

14

d.) Which sentence correctly states the result **with units**? (1)

(D) Between $x = 2$ and $x = 6$ hours, the distance driven changes at an average rate of 14 miles per hour.

WORKED SOLUTION

$$f(2) = 21, \quad f(6) = 77$$

$$\frac{f(6) - f(2)}{6 - 2} = \frac{77 - (21)}{4} = \frac{56}{4} = 14 \text{ miles per hour}$$

It's an *average* over an interval (not an instant), and the units are miles per hour — miles per hour. The total change would be 56 miles.

11. The table shows the U.S. unemployment rate.

5 pts

Year	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026
Rate, %	4.6	5.1	5.6	5.4	6.1	6.5	5.9	5.6	5.7	6.0	6.0

Rounded answers (2+ decimal places) are accepted.

a.) Average rate of change from 2017 to 2020 (2)

$$\frac{1}{3}$$

b.) Average rate of change from 2023 to 2026 (2)

$$\frac{2}{15}$$

c.) The units of these answers are... (1)

(A) percentage points per year

WORKED SOLUTION

2017–2020:

$$\frac{6.1 - 5.1}{2020 - 2017} = \frac{1}{3} \approx 0.333$$

2023–2026:

$$\frac{6.0 - 5.6}{2026 - 2023} = \frac{0.4}{3} \approx 0.133$$

Units: output units per input unit → **percentage points per year.**

12. Let $f(x) = -x^2 + 4x - 5$. Find and simplify the difference quotient $\frac{f(x+h)-f(x)}{h}$, then use it 7 pts to fill in the table.

a.) Simplified difference quotient (in terms of x and h) (3)

$$-2x - h + 4$$

b.) $x = 2, h = 1$ (1)

$$-1$$

c.) $x = 2, h = 0.1$ (1)

$$-\frac{1}{10}$$

d.) $x = 2, h = 0.01$ (1)

$$-\frac{1}{100}$$

e.) As $h \rightarrow 0$, what value do these approach? (1)

$$0$$

WORKED SOLUTION

$$f(x+h) = -(x+h)^2 + 4(x+h) - 5 = -x^2 - 2xh - h^2 + 4x + 4h - 5$$

$$f(x+h) - f(x) = -2xh - h^2 + 4h$$

$$\frac{f(x+h) - f(x)}{h} = \frac{h(-2x - h + 4)}{h} = -2x - h + 4$$

Now substitute:

$$x = 2, h = 1 : -1$$

$$x = 2, h = 0.1 : -\frac{1}{10} = -0.1$$

$$x = 2, h = 0.01 : -\frac{1}{100} = -0.01$$

Letting $h \rightarrow 0$ leaves 0: the instantaneous rate of change (slope of the tangent line) at $x = 2$. That's exactly what section 1.4 formalizes.

13. Let $f(x) = \frac{1}{x}$. Use the limit definition $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h)-f(x)}{h}$ to find $f'(x)$. 8 pts

a.) $f'(x) =$ (3)
 $\frac{-1}{x^2}$

b.) $f'(1) =$ (2)
 -1

c.) Slope of the tangent line to f at $x = 2$ (2)
 $-\frac{1}{4}$

d.) What does $f'(1) = -1$ mean? (1)

(C) The instantaneous rate of change of f at $x = 1$ is -1 — the slope of the tangent line there.

WORKED SOLUTION

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{\frac{1}{x+h} - \frac{1}{x}}{h} = \lim_{h \rightarrow 0} \frac{\frac{1x-1(x+h)}{x(x+h)}}{h} \\ &= \lim_{h \rightarrow 0} \frac{-1h}{x(x+h)} \cdot \frac{1}{h} = \lim_{h \rightarrow 0} \frac{-1}{x(x+h)} = \frac{-1}{x^2} \end{aligned}$$

Then $f'(1) = -1$ and the slope of the tangent at $x = 2$ is $f'(2) = -\frac{1}{4}$.

$f'(1)$ is the *instantaneous* rate of change at one input — the slope of the tangent line, not the height of the graph and not an average over an interval.

14. (a) Suppose $f(1) = -3$ and $f'(1) = -1$. Write the equation of the tangent line to f at $x = 1$. 7 pts

(b) The tangent line to the graph of g at $x = -1$ is $y = -3x - 3$. Find $g(-1)$ and $g'(-1)$.

a.) (a) Tangent line (3)

$$y = -x - 2$$

b.) (b) $g(-1)$ (2)

$$0$$

c.) (b) $g'(-1)$ (2)

$$-3$$

WORKED SOLUTION

(a) Point $(1, -3)$, slope -1:

$$y + 3 = -(x - 1) \Rightarrow y = -x - 2$$

(b) The tangent line touches g at $x = -1$, so $g(-1)$ is the line's height there:

$-3(-1) - 3 = 0$. Its slope is the derivative: $g'(-1) = -3$.

15. Let $h(t) = -16t^2 + 160t$ be the height, in feet, of a rocket t seconds after launch.

6 pts

a.) $h'(t) =$ (3)
 $-32t + 160$

b.) $h'(2)$ (2)
 96

c.) Interpret $h'(2) = 96$. (1)
(A) At $t = 2$ seconds, the height of the rocket is increasing at a rate of 96 feet per second.

WORKED SOLUTION

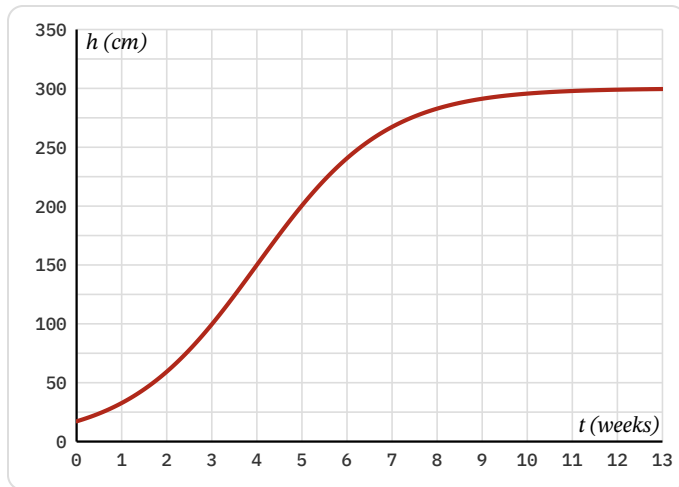
$$\begin{aligned} h'(t) &= \lim_{h \rightarrow 0} \frac{f(t+h) - f(t)}{h} = \lim_{h \rightarrow 0} \frac{[-16(t+h)^2 + 160(t+h)] - (-16t^2 + 160t)}{h} \\ &= \lim_{h \rightarrow 0} \frac{-16t^2 - 32th - 16h^2 + 160t + 160h + 16t^2 - 160t}{h} \\ &= \lim_{h \rightarrow 0} \frac{-32th - 16h^2 + 160h}{h} = \lim_{h \rightarrow 0} \frac{h(-32t - 16h + 160)}{h} \\ &= \lim_{h \rightarrow 0} (-32t - 16h + 160) = -32t + 160 \end{aligned}$$

$$h'(2) = -32(2) + 160 = 96$$

Positive/negative tells the direction, and the units are feet per second: at that instant the height of the rocket is increasing at 96 feet per second. (The value $h(2) = 256$ is a different quantity.)

16. A sunflower seedling is planted and its height h (in centimeters) after t weeks is graphed below. Decide whether each statement is true or false.

4 pts



- a.) The slopes of the tangent lines get steeper up to $t = 7$, and then the slopes start to flatten out. (1)
False
- b.) The average rate of change over $[0, 10]$ is greater than the instantaneous rate of change at $t = 3$. (1)
False
- c.) The seedling grows faster (instantaneous rate) at week 2 than at week 8. (1)
True
- d.) The average rate of change over $[10, 12]$ is less than over $[1, 3]$. (1)
True

WORKED SOLUTION

- a. **False.** The graph is steepest at about $t = 4$ (where it changes from bending up to bending down).
- b. **False.** $\text{ARC} \approx \frac{295.6 - 17.2}{10} \approx 27.8$ cm/week; the tangent slope at $t = 3$ is about 46.6 cm/week.
- c. **True.** Tangent slopes: about 33.3 cm/week at $t = 2$ vs. 11.3 at $t = 8$.
- d. **True.** $[10, 12]$: ≈ 1.7 cm/week; $[1, 3]$: ≈ 33.4 cm/week.
-